Лабораторна робота №1

№13.

$$\begin{cases} \frac{dx}{dt} = 2x - y \\ \frac{dy}{dt} = 5x - 2y \end{cases}$$

$$A = \begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix}$$

$$|A - \lambda E| = 0$$

$$\begin{vmatrix} 2-\lambda & -1 \\ 5 & -2-\lambda \end{vmatrix} = (2-\lambda)(-2-\lambda) + 5 = -(2-\lambda)(2-\lambda) + 5 =$$

$$= -(4 - \lambda^2) + 5 = \lambda^2 - 4 + 5 = \lambda^2 + 1 = 0$$

$$\lambda_1 = i$$

$$\lambda_2 = -i$$

$$\lambda_1 = i$$

$$\begin{pmatrix} 2-i & -1 \\ 5 & -2-i \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = 0$$

$$\begin{cases} (2-i)y_1 - y_2 = 0 \Rightarrow y_2 = (2-i)y_1 \\ 5y_1 + (-2-i)y_2 = 0 \Rightarrow \end{cases}$$

$$\Rightarrow 5y_1 - (2+i)(2-i)y_1 = 0$$

$$y = \begin{pmatrix} 1 \\ 2-i \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + i\begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$\bar{y}(t) = e^{\lambda_1 t} y = e^{it}\begin{pmatrix} 1 \\ 2-i \end{pmatrix}$$

$$\bar{y}(t) = (\cos t + i\sin t)\begin{pmatrix} 1 \\ 2-i \end{pmatrix} = \begin{pmatrix} \cos t + i\sin t \\ (2-i)(\cos t + i\sin t) \end{pmatrix} \Rightarrow$$

$$(2-i)(\cos t + i\sin t) = 2\cos t + 2i\sin t - i\cos t - i^2\sin t =$$

$$= (2\cos t + \sin t) + i(2\sin t - \cos t)$$

$$\Rightarrow \begin{pmatrix} \cos t \\ 2\cos t - \sin t \end{pmatrix} + i\begin{pmatrix} \sin t \\ 2\sin t - \cos t \end{pmatrix}$$

$$\overset{\downarrow\downarrow}{y^{(1)}(t)} \qquad \overset{\downarrow\downarrow}{y^{(2)}(t)}$$

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1\begin{pmatrix} \cos t \\ 2\cos t + \sin t \end{pmatrix} + C_2\begin{pmatrix} \sin t \\ 2\sin t - \cos t \end{pmatrix}$$

$$\boxed{\begin{cases} x(t) = C_1 \cdot \cos t + C_2 \cdot \sin t \\ y(t) = C_1 \cdot (2\cos t + \sin t) + C_2 \cdot (2\sin t - \cos t) \end{cases}}$$

$$\begin{cases} x(t) = C_1 \cdot \cos t + C_2 \cdot \sin t \\ y(t) = (2C_1 - C_2)\cos t + (C_1 + 2C_2)\sin t \end{cases}$$

$$t_0 = 0 \Rightarrow \begin{cases} x(0) = C_1 \\ y(0) = 2C_1 - C_2 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} C_1 = x_0 \\ C_2 = 2x_0 - y_0 \end{cases}$$

$$\begin{cases} x(t) = x_0 \cdot \cos t + (2x_0 - y_0) \cdot \sin t \\ y(t) = y_0 \cdot \cos t + (5x_0 - 2y_0) \cdot \sin t \end{cases}$$